



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD MPP.7

Math is Playful

Lesson 10-3 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

Language Objective: I can describe a puzzle strategy using the words pattern, rule, and predict.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

puzzle

pattern

pattern rule

predict

organize



Tap any word to see its meaning and a picture.

1

A problem designed to be played with — rules, a goal, room to imagine — is a

2

Objects or shapes repeated in a regular order form a

3

The rule for getting the next value from the one before is the

4

To say what will happen before it happens, using a pattern, is to

5

To arrange information so patterns become easier to see is to

 it.

Write About the Math

How can a pattern help you imagine the solution?

¿Cómo te puede ayudar un patrón a imaginar la solución?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **puzzle**
rompecabezas

☐ **pattern**
patrón

☐ **pattern rule**
regla del patrón

☐ **predict**
predecir

☐ **organize**
organizar

Model: The pattern rule is to double the previous number of moves and add 1: $1 \times 2 + 1 = 3$, and $3 \times 2 + 1 = 7$. — A strong answer states 'double and add 1' and checks it against both jumps ($1 \rightarrow 3$ and $3 \rightarrow 7$). A weak answer only notices the numbers are growing without naming the rule.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The pattern rule for the Tower of Hanoi is to ____ the number of steps for the smaller tower and ____ for each new disc. Applying this rule twice starting from 7 steps for a three-disc tower gives a five-disc tower a minimum of ____ steps.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- My evidence is ____, so ____.
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** puzzle
2. **Notes 2:** pattern
3. **Notes 3:** pattern rule
4. **Notes 4:** predict
5. **Notes 5:** organize
6. **Watch for this mistake:** *Doubling the previous number of steps but forgetting to add 1 — answering 14 instead of 15 for four discs, or 30 instead of 31 for five.*
7. **Try It 1:**
8. **Try It 2:**